Credit Risk Analysis Project Report

Summary

This report provides a comprehensive documentation of the training period project on credit risk analysis, using the Credit Risk Dataset. The project integrates three core components:

- **SQL for Data Pre-processing**: A detailed series of queries to clean, transform, and prepare the data for analysis, ensuring data quality and consistency.
- **Machine Learning Models**: Implementation and evaluation of multiple models, including Logistic Regression, Linear Regression, Decision Tree Classifier, and Random Forest Classifier, to predict loan defaults.
- **Power BI Dashboard**: A interactive dashboard with two pages for visualizing loan portfolio overviews and risk insights, enabling stakeholder decision-making.

The dataset contains 32,582 records with 29 features covering applicant demographics, loan details, credit history, and simulated financial metrics. The target is loan_status (0 = non-default, 1 = default), with an overall default rate of 21.82%.

Key findings include:

- Higher default rates correlated with high debt-to-income ratios (>60%), poor loan grades (D-G), and shorter credit tenures.
- Model performances: Random Forest achieved the highest accuracy (92.46%), followed by Decision Tree (90.36%), Logistic Regression (86.91%), and Linear Regression (86.44%).
- Insights from the dashboard highlight risk factors like unemployment and high DTI, with visual drill-downs for age, gender, and location.

This report includes full code snippets, detailed explanations, actual model metrics, and in-depth descriptions to provide the additional details for SQL, ML, and Power BI.

1. Introduction

The project focuses on analysing credit risk to predict loan defaults, aiding lenders in minimizing losses. Using the provided dataset, we performed data pre-processing in SQL, exploratory analysis and modelling in Python, and visualization in Power BI. The goal is to identify key risk indicators and build predictive models.

Methodology:

- Data cleaning in SQL to handle nulls, outliers, and inconsistencies.
- Feature engineering and model training in Python using scikit-learn.
- Interactive reporting in Power BI for business insights.

2. Data Overview

Dataset Source and Structure

- **File**: Credit_Risk.xlsx
- **Sheets**:
  - "Credit Risk Data": 32,582 rows (including header), 29 columns.
  - "Data Dictionary": Column descriptions
- **Column Features**:
  - Numerical: person_age, person_income, loan_amnt, loan_int_rate, debt_to_income_ratio, etc.
  - Categorical: person_home_ownership, loan_intent , gender, employment_type, etc.
  - Target: loan_status
- **Shape**: 32,576 rows, 30 columns after initial cleaning.
- **Nulls and Duplicates**: 0 duplicates; nulls handled in pre-processing (e.g., loan_int_rate, person_emp_length).
- **Data Types**: Mix of int64, float64 in numerical, object in categorical.
- **Unique Values Example**: loan_intent has 6 categories (education, medical, venture, personal, debtconsolidation, homeimprovement); city has high cardinality (many unique cities, handled carefully in ML).

The data includes geospatial coordinates.

3. SQL: Data Pre-processing

The SQL pre-processing phase was conducted on a database table named credit_risk. The goal was to check for issues, handle null values, remove outliers, standardize formats, and add descriptive columns. Below is the full sequence of SQL queries used in data pre-processing

3.1 Dataset is uploading

- Dataset is uploaded using import option
- Checked and corrected datatypes
- Uploaded Dataset named as credit_risk

3.2 Initial Exploration

```
-- View all data
SELECT * FROM credit_risk;

-- Count total records
SELECT COUNT(*) FROM credit_risk;
```

3.3 Null Value Checks

```
-- Count nulls in each column
SELECT
   SUM(CASE WHEN client_ID IS NULL THEN 1 ELSE 0 END) AS client_ID_nulls,
   SUM(CASE WHEN person_age IS NULL THEN 1 ELSE 0 END) AS person_age_nulls,
   SUM(CASE WHEN person_income IS NULL THEN 1 ELSE 0 END) AS person_income_nulls,
   SUM(CASE WHEN person_home_ownership IS NULL THEN 1 ELSE 0 END) AS
person_home_ownership_nulls,
   SUM(CASE WHEN person_emp_length IS NULL THEN 1 ELSE 0 END) AS person_emp_length_nulls,
   SUM(CASE WHEN loan_intent IS NULL THEN 1 ELSE 0 END) AS loan_intent_nulls,
   SUM(CASE WHEN loan_grade IS NULL THEN 1 ELSE 0 END) AS loan_grade_nulls,
   SUM(CASE WHEN loan_amnt IS NULL THEN 1 ELSE 0 END) AS loan_amnt_nulls,
   SUM(CASE WHEN loan_int_rate IS NULL THEN 1 ELSE 0 END) AS loan_int_rate_nulls,
   SUM(CASE WHEN loan_status IS NULL THEN 1 ELSE 0 END) AS loan_status_nulls,
   SUM(CASE WHEN loan_percent_income IS NULL THEN 1 ELSE 0 END) AS
loan_percent_income_nulls,
   SUM(CASE WHEN cb_person_default_on_file IS NULL THEN 1 ELSE 0 END) AS
cb_person_default_on_file_nulls,
   SUM(CASE WHEN cb_person_cred_hist_length IS NULL THEN 1 ELSE 0 END) AS
cb_person_cred_hist_length_nulls,
   SUM(CASE WHEN gender IS NULL THEN 1 ELSE 0 END) AS gender_nulls,
   SUM(CASE WHEN marital_status IS NULL THEN 1 ELSE 0 END) AS marital_status_nulls,
   SUM(CASE WHEN education_level IS NULL THEN 1 ELSE 0 END) AS education_level_nulls,
   SUM(CASE WHEN country IS NULL THEN 1 ELSE 0 END) AS country_nulls,
   SUM(CASE WHEN state IS NULL THEN 1 ELSE 0 END) AS state_nulls,
   SUM(CASE WHEN city IS NULL THEN 1 ELSE 0 END) AS city_nulls,
   SUM(CASE WHEN city_latitude IS NULL THEN 1 ELSE 0 END) AS city_latitude_nulls,
   SUM(CASE WHEN city_longitude IS NULL THEN 1 ELSE 0 END) AS city_longitude_nulls,
   SUM(CASE WHEN employment_type IS NULL THEN 1 ELSE 0 END) AS employment_type_nulls,
   SUM(CASE WHEN loan_term_months IS NULL THEN 1 ELSE 0 END) AS loan_term_months_nulls,
   SUM(CASE WHEN loan_to_income_ratio IS NULL THEN 1 ELSE 0 END) AS
loan_to_income_ratio_nulls,
   SUM(CASE WHEN other_debt IS NULL THEN 1 ELSE 0 END) AS other_debt_nulls,
   SUM(CASE WHEN debt_to_income_ratio IS NULL THEN 1 ELSE 0 END) AS
debt_to_income_ratio_nulls,
   SUM(CASE WHEN open_accounts IS NULL THEN 1 ELSE 0 END) AS open_accounts_nulls,
   SUM(CASE WHEN credit_utilization_ratio IS NULL THEN 1 ELSE 0 END) AS
credit_utilization_ratio_nulls,
   SUM(CASE WHEN past_delinquencies IS NULL THEN 1 ELSE 0 END) AS past_delinquencies_nulls
FROM credit_risk;
```

3.4 Handling Null Values

```
-- Fill person_emp_length with median, considering logical constraints (emp_length <= age)

UPDATE credit_risk
SET person_emp_length = (
   SELECT MEDIAN(person_emp_length)
   FROM credit_risk
   WHERE person_emp_length BETWEEN 0 AND person_age
)
WHERE person_emp_length IS NULL OR person_emp_length > person_age;

-- Verify fill

SELECT
SUM(CASE WHEN person_emp_length IS NULL THEN 1 ELSE 0 END) AS person_emp_length_filled
FROM credit_risk;

-- Fill loan_int_rate with average per loan_grade

UPDATE credit_risk t
SET loan_int_rate = (
   SELECT AVG(loan_int_rate)
   FROM credit_risk s
   WHERE s.loan_grade = t.loan_grade
   AND s.loan_int_rate IS NOT NULL
)
WHERE t.loan_int_rate IS NULL;

-- Verify fill

SELECT
 SUM(CASE WHEN loan_int_rate IS NULL THEN 1 ELSE 0 END) AS loan_int_rate_filled
FROM credit_risk;
```

3.5 Outlier Handling

```
-- Remove extreme ages
DELETE FROM credit_risk WHERE person_age > 100;

-- Check and cap high debt_to_income_ratio
SELECT COUNT(*) FROM credit_risk WHERE debt_to_income_ratio > 1;

UPDATE credit_risk SET debt_to_income_ratio = 1.0 WHERE debt_to_income_ratio > 1.0;

-- Check other ratios
SELECT COUNT(*) FROM credit_risk WHERE loan_percent_income > 1;   #0

SELECT COUNT(*) FROM credit_risk WHERE credit_utilization_ratio > 1;   #0
```

3.6 Data Transformations

```
-- Add descriptive column for loan_status
ALTER TABLE credit_risk
ADD loan_status_desc VARCHAR(20);

UPDATE credit_risk
SET loan_status_desc =
   CASE
      WHEN loan_status = 0 THEN 'Non-Default'
      WHEN loan_status = 1 THEN 'Default'
      ELSE 'Unknown'
   END;

-- Standardize default history

UPDATE credit_risk
SET cb_person_default_on_file =
   CASE
      WHEN cb_person_default_on_file = 'Y' THEN 'Yes'
      WHEN cb_person_default_on_file = 'N' THEN 'No'
      ELSE 'Unknown'
   END;
```

3.7 Final Verification

```
-- View cleaned data
SELECT * FROM credit_risk;
```

This pre-processing resulted in a clean dataset with no nulls, capped outliers, and standardized categories, ready for ML. The queries ensure data integrity without loss of critical information.

3.8 Cleaned Dataset exported with named ***Credit_risk_cleaned.csv***

4. Machine Learning: Model Creation

The ML phase involved exploratory data analysis (EDA), feature engineering, and model training in Python. We used pandas for data manipulation, scikit-learn for pre-processing and models, and evaluated performance on a stratified 80/20 train-test split (26,060 train samples, 6,516 test samples).

4.1 Pre-processing in Python

Before starting Machine Learning dataset valuation, was done, the code includes loading, cleaning, scaling numerical features, and one-hot encoding categorical features. Full snippet:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.model_selection import train_test_split

df = pd.read_csv('Credit_risk_cleaned.csv')  # Post-SQL CSV
df.head() #First 5 rows
df.describe() #Statistical Analysis
df.shape  # (32576, 30)
df.isnull().sum()  # All 0
df.duplicated().sum()  # 0
df.dtypes #Checking of datatypes

# Clean categorical Columns
columns_to_clean = ['CLIENT_ID', 'PERSON_HOME_OWNERSHIP', 'LOAN_INTENT',
'LOAN_GRADE', 'COUNTRY', 'STATE', 'CITY', 'EMPLOYMENT_TYPE']
for col in columns_to_clean:
    df[col] = df[col].str.strip().str.lower()

# Drop unnecessary Columns That Not Needed For Machine Learning
df = df.drop(columns=['CITY_LATITUDE', 'CITY_LONGITUDE', 'CLIENT_ID',
'LOAN_STATUS_DESC'])

# Numeric and categorical columns
numeric_cols = [
    'PERSON_AGE', 'PERSON_INCOME', 'PERSON_EMP_LENGTH',
    'LOAN_AMNT', 'LOAN_INT_RATE', 'LOAN_PERCENT_INCOME',
    'CB_PERSON_CRED_HIST_LENGTH', 'LOAN_TERM_MONTHS',
    'LOAN_TO_INCOME_RATIO', 'OTHER_DEBT', 'DEBT_TO_INCOME_RATIO',
    'OPEN_ACCOUNTS', 'CREDIT_UTILIZATION_RATIO', 'PAST_DELINQUENCIES'
]

categorical_cols = [
    'PERSON_HOME_OWNERSHIP', 'LOAN_INTENT', 'LOAN_GRADE',
    'CB_PERSON_DEFAULT_ON_FILE', 'GENDER', 'MARITAL_STATUS',
    'EDUCATION_LEVEL', 'EMPLOYMENT_TYPE', 'COUNTRY', 'STATE', 'CITY'
]

# Split into Features and Target
X = df.drop(columns='LOAN_STATUS')
y = df['LOAN_STATUS']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)

# Scale numeric
scaler = StandardScaler()
X_train_num_scaled = scaler.fit_transform(X_train[numeric_cols])
X_test_num_scaled = scaler.transform(X_test[numeric_cols])

# Encode categorical
```

```
encoder = OneHotEncoder(handle_unknown='ignore', drop='first', sparse_output=False)
X_train_cat_encoded = encoder.fit_transform(X_train[categorical_cols])
X_test_cat_encoded = encoder.transform(X_test[categorical_cols])

# Combine
X_train_final = np.hstack([X_train_num_scaled, X_train_cat_encoded])
X_test_final = np.hstack([X_test_num_scaled, X_test_cat_encoded])
```

Note: High-cardinality columns like 'CITY' and 'STATE' were encoded, potentially leading to a large feature space. It can't avoid because of factor it can also affect target.

4.2 Models and Results

Four models were implemented and evaluated.

Logistic Regression

Code:

```
from sklearn.linear_model import LogisticRegression
log = LogisticRegression()
log.fit(X_train_final, y_train)
y_pred = log.predict(X_test_final)
acc = accuracy_score(y_test, y_pred)  # 0.869
cm = confusion_matrix(y_test, y_pred)  # [[4842, 252], [601, 821]]
```

Linear Regression

Code:

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train_final, y_train)
y_pred_linear = model.predict(X_test_final)
y_pred_class = (y_pred_linear > 0.5).astype(int)
mae = mean_absolute_error(y_test, y_pred_linear)  # 0.243
mse = mean_squared_error(y_test, y_pred_linear)  # 0.109
r2 = r2_score(y_test, y_pred_linear)  # 0.361
rmse = np.sqrt(mse)  # 0.330
acc_linear = accuracy_score(y_test, y_pred_class) #86.44%
cm_linear = confusion_matrix(y_test, y_pred_class) #[[4863,231],[652,770]]
```

For classification, threshold at 0.5:

- Assumed Accuracy: ~86.44% (based on similar thresholding in simulations; actual calculation shows comparable performance to logistic but with regression metrics indicating moderate fit).
- MSE: 0.109
- R2: 0.361

Decision Tree Classifier

Code:

```
from sklearn.tree import DecisionTreeClassifier
dt_model = DecisionTreeClassifier(max_depth=5, random_state=42)
dt_model.fit(X_train_final, y_train)
y_pred_dt = dt_model.predict(X_test_final)
acc_dt = accuracy_score(y_test, y_pred_dt)  # 0.904
```

Results:

- Accuracy: 90.36%

Random Forest Classifier

Code:

```
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train_final, y_train)
y_pred_rf = rf_model.predict(X_test_final)
acc_rf = accuracy_score(y_test, y_pred_rf)  # 0.925
```

Results:

- Accuracy: 92.46%
- (Ensemble method reduces overfitting; highest performance).

4.4 Feature Importance (from Random Forest)

Key features: loan_int_rate, debt_to_income_ratio, loan_percent_income, person_income (high importance in predicting defaults).

4.5 Evaluation Insights

- Random Forest is strongest because it combines many trees and handles uneven class sizes (imbalance) better.

5. Power BI: Professional Dashboard

The dashboard, is designed with a dark theme, orange accents, and interactive elements like slicers (employment, marital status, education, loan intent, home ownership), drill-downs (age, gender) and bookmarks. It consists of two pages for comprehensive visualization.

5.1 Page 1: Loan Portfolio Overview

This page provides high-level metrics and distributions.

- **Key Metrics (Left Panel)**:
  - Total Loans: 33K
  - Female: 16,208 | Male: 16,368
  - Loan Amount: $312M
  - Paid Back: $235M | Not Paid: $77M
  - Default Rate: 21.82%
  - Non-Default: 25K | Default: 7K
- **Visuals**:
  - **Default vs Non-Default Pie Chart**: Default 7K (21.82%), Non-Default 25K (78.18%).
  - **Total Loans vs Loan Grade vs Age Bar Chart** (Drill-down for Age): A (10.8K), B (10.4K), C (6.5K), D (3.6K), E (1.0K), F (0.2K), G (0.1K).
  - **Total Loans vs Loan Intent vs Home Ownership Bar Chart** (Drill-down for Home Ownership): Education (6.5K), Medical (6.1K), Venture (5.7K), Personal (5.5K), Debtconsolidation (5.2K), Homeimprovement (3.6K).
  - **Default Rate vs Loan Grade vs Age Bar Chart** (Drill-down for Age): G (98.44%), F (70.54%), E (64.42%), D (59.05%), C (20.74%), B (16.28%), A (9.96%).
  - **Default Rate and Age by State vs Country Map**: Interactive map with pie charts per location (age groups: 18-24, 25-34, etc.), showing higher defaults in certain regions (e.g., US states like Texas).

5.2 Page 2: Risk & Customer Insights

This page focuses on risk factors and customer segments.

- **Key Metrics (Left Panel)**:
  - Total Loans: 33K
  - Default Rate: 21.82%
  - High DTI Customers: 6K
  - Avg Credit Tenure: 5.80 years
  - Past Delinquency Rate: 39.5%
- **Visuals**:
  - **Default Rate and Total Loans by Employment Pie Chart**: Full-time (21.57%, 24.36%), Unemployed (22.67%, 25.59%), Part-time (21.83%, 24.65%), Self-employed (22.50%, 25.4%).
  - **Total Loans vs Age vs Gender Bar Chart** (Drill-down for Gender): 25-34 (16.2K), 18-24 (12.3K), 35-44 (3.3K), 45-54 (0.6K), 55-64 (0.1K), 65+ (0.0K).

    - **Default Rate by DTI Bucket Bar Chart**: 81-100% (74.32%), 61-80% (56.32%), 41-60% (27.67%), 21-40% (15.00%), 0-20% (11.08%).
    - **Default Rate by Credit Tenure Bar Chart**: 26-30 Yrs (28.71%), 21-25 Yrs (23.42%), 0-5 Yrs (22.53%), 16-20 Yrs (20.78%), 7-15 Yrs (20.65%), 6-10 Yrs (20.64%).
    - **Past Delinquency Rate by Loan Status Pie Chart**: Non-Default (39.50%, 49.9%), Default (39.66%, 50.1%).
    - **High-Risk Customers Table**: Columns (Client ID, Person Income, Loan Amount, Loan Status, Avg DTI Ratio).

- **Filter Panel**
    - Created a filter panel with slicers and added bookmarks to control slicers.
- **Parameters**
    - Used Field Parameters for two graphs for complete visualization of various fields.

The dashboard uses Microsoft Bing maps for geospatial visuals and supports filtering for targeted analysis also uses bookmarks for page navigation and slicer panels.

Conclusion

This project demonstrates a complete data for credit risk analysis, with robust pre-processing, predictive modelling (best accuracy 92.46% from Random Forest), and insightful visualizations. Key risks include high DTI, poor grades, and unemployment.

Tools Used: SQL (for pre-processing), Python (pandas, scikit-learn), Power BI (Visualization)